WEEK 6 · VECTOR AUTOREGRESSION

VAR, Cholesky, IRF and FEVD

Interest rates and stock returns, calculated step by step

Granger causality · Cholesky decomposition · impulse responses · variance decomposition

Every coefficient in this deck is illustrative and chosen for exact hand calculation.
They are not empirical estimates of the relationship between interest rates and stock returns.

One fitted VAR answers three different questions

The tools share the same model, but each asks something different

Tool	Question it answers	What it needs
Granger causality	Does past information improve prediction?	Lag coefficients of the VAR
IRF	How does each variable respond to one individual shock?	A and the impact matrix P
FEVD	Which shocks contribute to forecast uncertainty?	Squared IRFs, accumulated

VAR

Predict both variables jointly

Forecast errors

u: what the VAR did not predict

Granger test

Separate question on lags

Cholesky

Turn u into uncorrelated ε

IRF

Response to one individual shock

FEVD

Allocate forecast error variance

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PART 1 · VAR

Vector Autoregression and Forecast Errors

Two equations, one lag, and the part of today that yesterday could not predict

A VAR models several variables with each other's lags

Vector autoregression (VAR): the multivariate extension of an autoregressive (AR) model

A VAR(p) regresses every variable on p lags of itself and p lags of every other variable in the system.

$$y_t = c + A_1 y_{t-1} + \cdots + A_p\, y_{t-p} + u_t, \qquad \mathrm{E}[u_t] = 0, \quad \mathrm{E}[u_t u_t'] = \Sigma_u, \quad \mathrm{E}[u_t u_\tau'] = 0 \ (t \neq \tau)$$

Why use a VAR

- Rates and returns can feed back on each other
- One model gives joint forecasts and shock paths

What is assumed about u

- Mean zero with covariance matrix Σu
- Uncorrelated across time, possibly correlated within a period

Six terms recur throughout the VAR calculations

Each term is defined here before it is used

Term	Definition
Lag	The value of a variable k periods earlier; s_{t-1} is the one-period lag
Stationarity	Mean, variance and autocovariances do not change over time
Fitted value $\hat{y}_t$	The model's prediction of y_t from past information
Residual $\hat{u}_t$	Actual minus fitted, $\hat{u}_t = y_t - \hat{y}_t$; the sample estimate of u_t
Forecast error u_t	Also called the innovation: the part of y_t past information cannot predict
Cointegration · VECM	Nonstationary levels with a stationary combination; a VECM adds an error-correction term

Each equation uses yesterday's values of both

r_t is the interest rate, s_t is the stock return, and the model uses one lag

$$r_t = c_r + a_{11}\, r_{t-1} + a_{12}\, s_{t-1} + u_{r,t}$$
$$s_t = c_s + a_{21}\, r_{t-1} + a_{22}\, s_{t-1} + u_{s,t}$$

Reading the rate equation

- Constant c_r
- Contribution of yesterday's rate
- Contribution of yesterday's stock return
- Unexpected component u_r,t

The return equation works the same way

- Same past information: r_{t-1} and s_{t-1}
- Its own constant and coefficients
- u is the part the past cannot explain

Matrix notation packages the two equations into one

y_t stacks r_t and s_t into a column vector; A collects the four lag coefficients

$$\underbrace{\begin{pmatrix} r_t \\ s_t \end{pmatrix}}_{y_t} = \underbrace{\begin{pmatrix} c_r \\ c_s \end{pmatrix}}_{c} + \underbrace{\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} r_{t-1} \\ s_{t-1} \end{pmatrix}}_{y_{t-1}} + \underbrace{\begin{pmatrix} u_{r,t} \\ u_{s,t} \end{pmatrix}}_{u_t}$$

Symbol	Meaning
t and t−1	Current and previous periods
c	Vector of constants
A	2 × 2 matrix of lagged effects
u_t	Forecast error vector: what past information did not predict

VAR(1) is a modelling choice that must be checked

VAR(1) means one lag; several lags are possible and are chosen from the data

Stationarity

Check that the variables are stationary before estimating

Lag length

One lag here; in practice choose the number of lags

Residuals

Inspect residual behaviour after fitting

Returns vs levels

Stock returns are often used instead of price levels; cointegrated levels may call for a VECM

All numbers that follow come from an illustrative VAR(1), not estimation.

The fitted VAR separates prediction from surprise

A fitted VAR has coefficients estimated from data; hats mark estimated quantities

$$\hat{y}_t = \hat{c} + \hat{A}\, y_{t-1} \qquad \hat{u}_t = y_t - \hat{y}_t$$

Variable	Predicted	Actual	Error
Interest rate	3.0%	3.2%	+0.2 pp
Stock return	+0.5%	−1.5%	−2.0 pp

The rate was expected to be 3.0%, so the surprise is 0.2 percentage points, not the whole observed 3.2%.

Reading the residual

- Fitted value: prediction from past information
- Residual: actual minus fitted
- Residuals estimate the forecast error properties

u and ε are two different kinds of shocks

The distinction between them is the backbone of the rest of the deck

u — reduced-form forecast errors

- Come directly from the VAR
- Unpredictable from past information
- Can still be correlated with the other variable's error in the same period

ε — orthogonalized shocks

- Constructed later from u
- Uncorrelated with each other
- Each has variance one

For simplicity, hats are omitted in the remaining calculations.

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PART 2 · GRANGER

Granger Causality and Actual Causation

Predictive content is a statement about forecasts, not about mechanisms

Granger causality is defined by forecast improvement

Clive Granger (1969): x Granger-causes y if past x helps predict y beyond past y alone

r Granger-causes s if adding past r lowers the forecast error of s, given that past s is already used.

$$\mathrm{MSE}\left(\hat{s}_t \mid s_{t-1}, s_{t-2}, \ldots, r_{t-1}, r_{t-2}, \ldots\right) < \mathrm{MSE}\left(\hat{s}_t \mid s_{t-1}, s_{t-2}, \ldots\right)$$

Null hypothesis H_0

- The default claim tested: all lagged r coefficients are zero
- Rejecting H_0: past r adds predictive power

Test statistic

- Wald or F test on the lagged r coefficients
- One joint test when there are several lags

Granger asks whether past rates improve forecasts

Does adding past interest rates help beyond what past stock returns already tell us?

$$s_t = c_s + a_{21}\, r_{t-1} + a_{22}\, s_{t-1} + u_{s,t} \qquad H_0 :\ a_{21} = 0$$

How the test is read

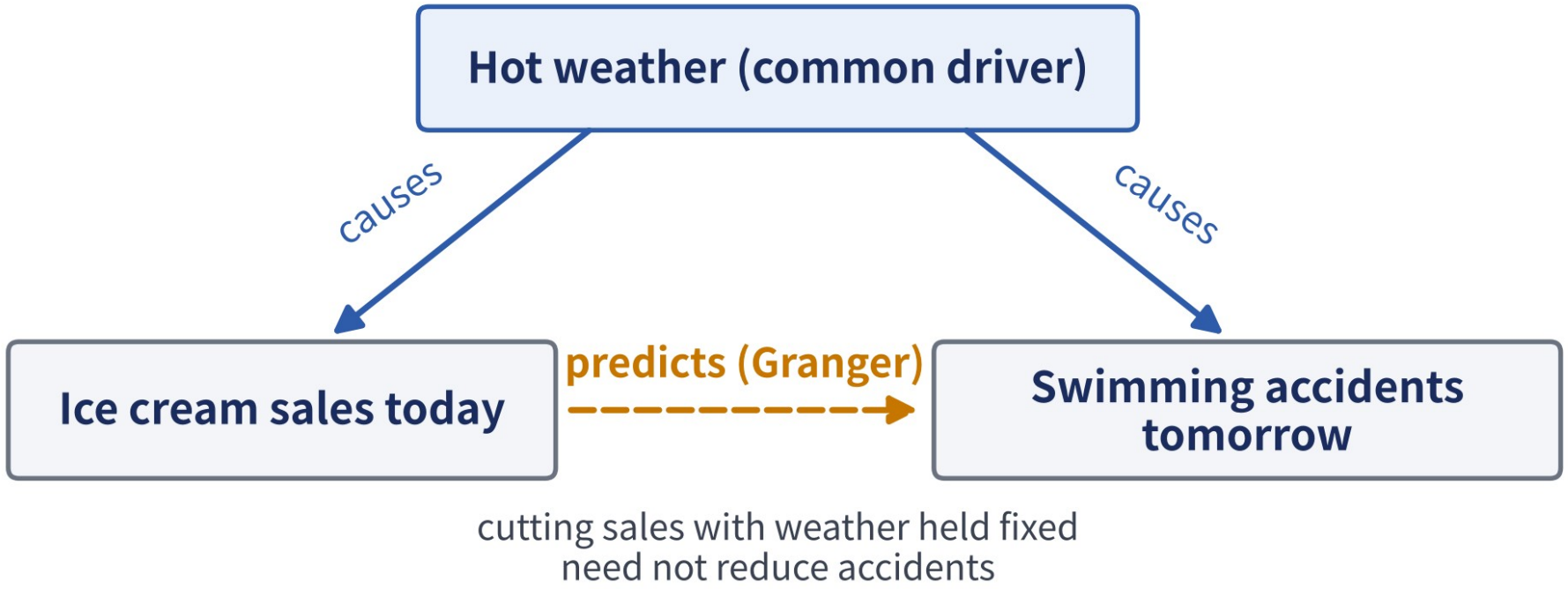
- In a VAR(1), rejecting H_0 supports "rates Granger-cause returns"
- With several lags, test all lagged rate coefficients jointly
- The result depends on the variables and lags included

What the test does not need

- No Cholesky decomposition
- Not a gate that must be passed before IRF or FEVD
- A separate question answered with the same VAR

An early signal need not be a cause

Rates and stocks may react to the same news at different speeds; precedence alone is not causation



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PART 3 · CHOLESKY

Orthogonalizing the Forecast Errors

Building uncorrelated, unit-variance shocks from correlated errors

Covariance and correlation measure co-movement

Definitions needed before uncorrelated shocks can be built

$$\mathrm{Var}(X) = \mathrm{E}\big[(X-\mu_X)^2\big] \qquad \mathrm{Cov}(X,Y) = \mathrm{E}\big[(X-\mu_X)(Y-\mu_Y)\big] \qquad \mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\,\sigma_Y}$$

Term	Definition
Variance	Average squared deviation from the mean; σ = $\sqrt{}$ variance
Covariance	Average product of joint deviations; sign gives direction
Correlation	Covariance scaled to lie between −1 and 1
Covariance matrix Σ	Variances on the diagonal, covariances off it; symmetric
Orthogonal (uncorrelated)	Covariance equal to zero
Independent	One says nothing about the other; stronger than uncorrelated

Common news makes forecast errors move together

Example: an unexpected inflation announcement moves both rates and stock returns

$$\mathrm{Cov}(u_{r,t},\, u_{s,t}) \neq 0 \qquad \Sigma_u = \begin{pmatrix} \sigma_r^2 & \sigma_{rs} \\ \sigma_{rs} & \sigma_s^2 \end{pmatrix}$$

Row / column	Rate error	Return error
Rate error	σ_r^2	σ_{rs}
Return error	σ_{rs}	σ_s^2

Two variances and one distinct covariance; the off-diagonal entries are equal and are estimated from residuals.

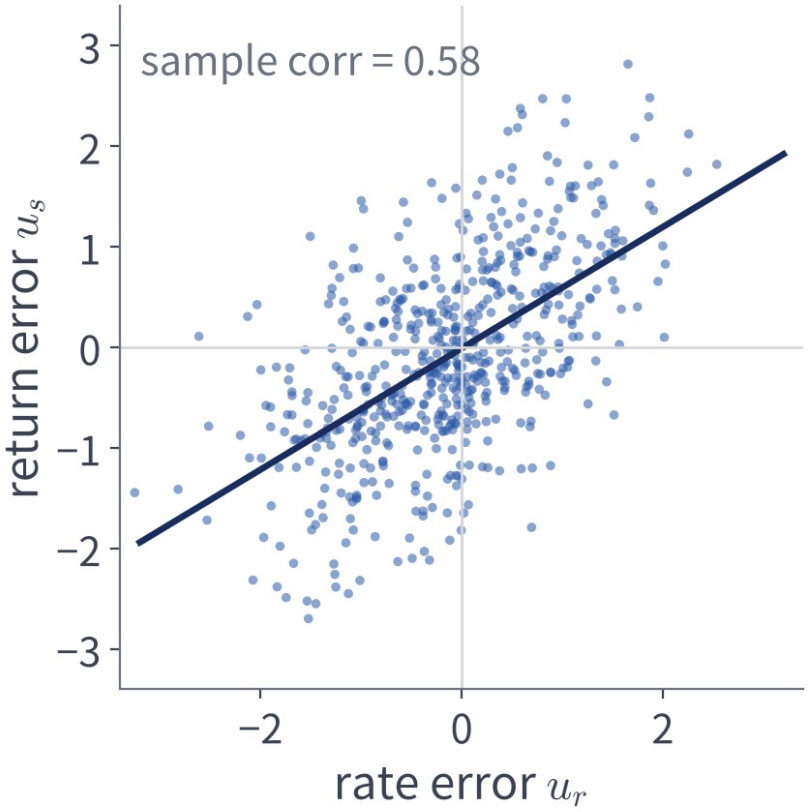
Why create new shocks

- IRF: trace one shock at a time
- FEVD: allocate variance across shocks
- Overlap makes the allocation ambiguous
- Cholesky is one way out, not needed for every IRF

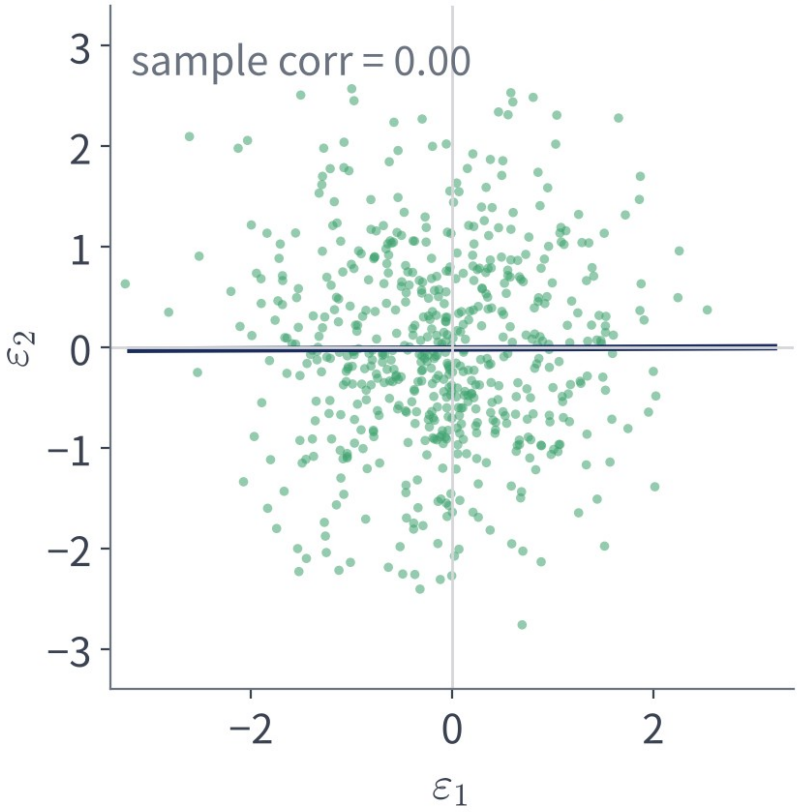
Orthogonalization removes the overlap

Simulated errors with correlation 0.6, before and after the transformation $\varepsilon = P^{-1}u$

Original forecast errors u (corr = 0.6)



New shocks $\varepsilon = P^{-1}u$ (corr = 0)



New shocks: a triangular mix plus two conditions

Order the interest rate first and the stock return second

$$u_r = p_{11}\,\varepsilon_1$$
$$u_s = p_{21}\,\varepsilon_1 + p_{22}\,\varepsilon_2$$

Mixing equations

$$\mathrm{Var}(\varepsilon_1) = \mathrm{Var}(\varepsilon_2) = 1 \qquad \mathrm{Cov}(\varepsilon_1, \varepsilon_2) = 0$$

Conditions stated separately: part of the definition

Variance one

- Standardizes shock size

Covariance zero

- Orthogonalizes the shocks

Caution

- Uncorrelated is not the same as statistically independent

Six matrix terms make the Cholesky step readable

Shown for a 2 × 2 matrix, the size used in this deck

$$L = \begin{pmatrix} \ell_{11} & 0 \\ \ell_{21} & \ell_{22} \end{pmatrix} \qquad L' = \begin{pmatrix} \ell_{11} & \ell_{21} \\ 0 & \ell_{22} \end{pmatrix} \qquad I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad x'\Sigma x > 0 \ \text{ for all } x \neq 0$$

Term	Definition
Lower triangular matrix	All entries above the diagonal are zero
Transpose L′	Rows and columns swapped
Identity matrix I	Ones on the diagonal, zeros elsewhere; the matrix version of 1
Inverse P^{-1}	The matrix that undoes P: $P^{-1}P = I$
Positive definite	$x'\Sigma x > 0$ for every nonzero x: all combinations have positive variance
Cholesky decomposition	Writing positive definite Σ as LL′, L lower triangular

Cholesky decomposition factors Σu into PP′

P is lower triangular; the prime denotes the transpose

$$u = P\varepsilon, \qquad P = \begin{pmatrix} p_{11} & 0 \\ p_{21} & p_{22} \end{pmatrix}, \qquad \mathrm{Var}(\varepsilon) = I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Sigma_u = \mathrm{Var}(P\varepsilon) = P\,I\,P' = PP'$$

Uniqueness

- Σu positive definite → P unique for a given ordering when its diagonal is positive

Positive definiteness

- Every nonzero combination of the errors has positive variance; a singular Σu can block P^{-1}

Cholesky strips the overlap, then rescales

Rate first: ε_1 is the standardized rate error; ε_2 is what is left of the return error

$$\varepsilon_1 = \frac{u_r}{\sigma_r} \qquad v = u_s - \frac{\sigma_{rs}}{\sigma_r^2}\, u_r \qquad \varepsilon_2 = \frac{v}{\mathrm{SD}(v)}$$

1 · Standardize u_r

ε_1 is the rate error measured in standard deviations

$p_{11} = \sigma_r$

2 · Strip the overlap

Regress u_s on u_r; the residual v shares nothing with u_r

$p_{21} = \sigma_{rs} / \sigma_r$

3 · Rescale

Divide v by its standard deviation so that $\mathrm{Var}(\varepsilon_2) = 1$

$p_{22} = SD(v)$

Correlation 0.6 gives $p_{21} = 0.6$ and $p_{22} = 0.8$

Both error variances equal one in this example; the full step-by-step check is in Appendix A

$$\Sigma_u = \begin{pmatrix} 1 & 0.6 \\ 0.6 & 1 \end{pmatrix} \Rightarrow P = \begin{pmatrix} 1 & 0 \\ 0.6 & 0.8 \end{pmatrix} \qquad u_r = \varepsilon_1, \quad u_s = 0.6\,\varepsilon_1 + 0.8\,\varepsilon_2$$

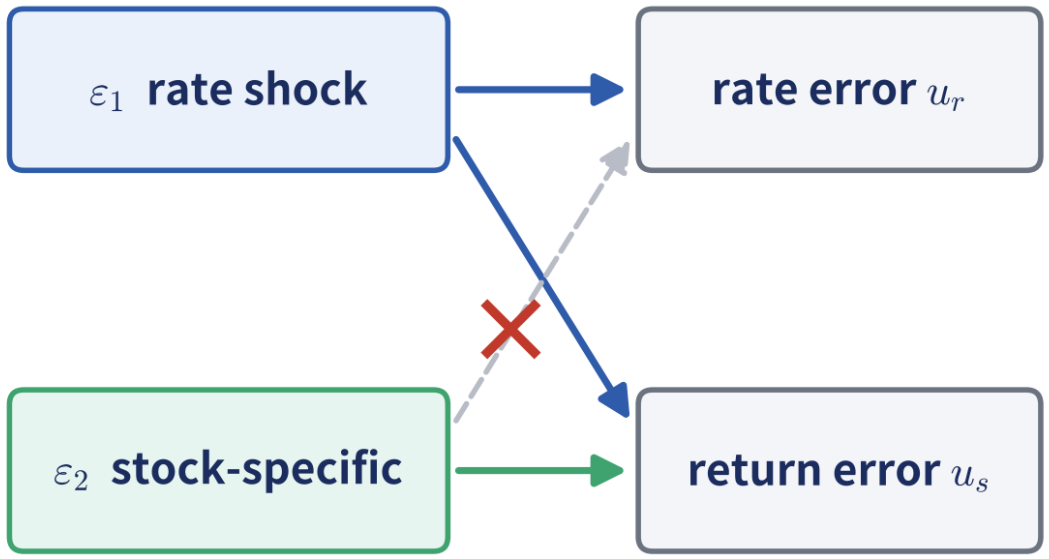
$0.6\varepsilon_1$ is the part shared with the rate error; $0.8\varepsilon_2$ is the rest. Check: $0.6^2 + 0.8^2 = 1 = \mathrm{Var}(u_s)$.

Uncorrelated shocks are constructed, not assumed.

The ordering sets the same-period restrictions

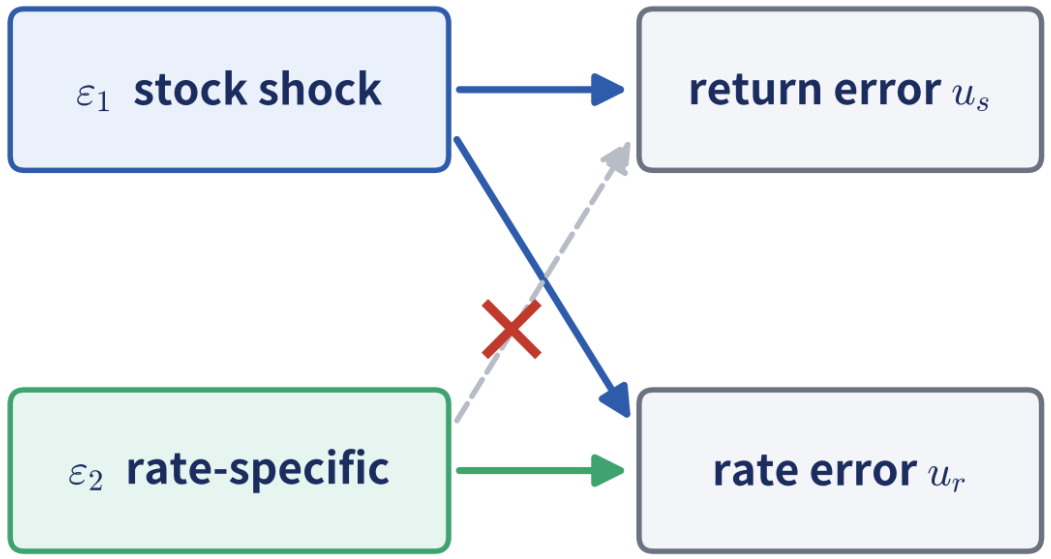
With rates first, ε_2 does not enter the contemporaneous rate error equation

Ordering: rate → return



no same-period arrow from ε_2 to the first variable

Ordering: return → rate



no same-period arrow from ε_2 to the first variable

Interpretation inherits the ordering's restrictions

Naming ε_1 a rate shock and ε_2 stock-specific accepts these restrictions

Ordering	Contemporaneous restriction
Rate → return	Stock-specific shock has zero impact on rates
Return → rate	Rate-specific shock has zero impact on returns

Same period depends on frequency

- Same day for daily data, same month for monthly data
- From the next period, effects run both ways

What the data cannot settle

- Both orderings compute; results can differ
- Orthogonalization alone is not causation

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PART 4 · IRF

Impulse Response Functions

Impact, one period later, and two periods later

An IRF is the response to one individual shock

Impulse response function: each variable's path after a single shock, all other shocks held at zero

$\psi_{i\,j,h}$ is how much variable i differs h periods later because shock j alone took the value one today.

$$\psi_{ij,h} = \frac{\partial\, y_{i,t+h}}{\partial\, \varepsilon_{j,t}} \qquad \Psi_h = A^h P \;\; (\mathrm{VAR}(1)), \qquad h = 0, 1, 2, \ldots$$

Why the shock must be individual

- Correlated u cannot move alone; orthogonal ε can

Reading terms

- Impact h = 0; horizon h = periods later
- One-SD shock: Var(ε) = 1; cumulative = sum of responses

A simple VAR combines lag matrix A with P

A propagates past values; P carries today's shocks into today's errors

$$A = \begin{pmatrix} 0.5 & 0 \\ -0.2 & 0.4 \end{pmatrix} \qquad P = \begin{pmatrix} 1 & 0 \\ 0.6 & 0.8 \end{pmatrix}$$

$$r_t = c_r + 0.5\, r_{t-1} + \varepsilon_{1,t}$$
$$s_t = c_s - 0.2\, r_{t-1} + 0.4\, s_{t-1} + 0.6\, \varepsilon_{1,t} + 0.8\, \varepsilon_{2,t}$$

Units

- Illustrative coefficients; responses are in the model's variable units, not calibrated market units

An IRF moves one shock and holds the other at zero

Δ is the difference between the path with that single shock and the path without it

$$\Delta r_{\text{next}} = 0.5\,\Delta r_{\text{current}}$$
$$\Delta s_{\text{next}} = -0.2\,\Delta r_{\text{current}} + 0.4\,\Delta s_{\text{current}}$$

$$\text{Only } \varepsilon_1 = 1: \quad \begin{pmatrix}\Delta r_t \\ \Delta s_t\end{pmatrix} = \begin{pmatrix}p_{11} \\ p_{21}\end{pmatrix} = \begin{pmatrix}1 \\ 0.6\end{pmatrix}$$

$$\text{Only } \varepsilon_2 = 1: \quad \begin{pmatrix}\Delta r_t \\ \Delta s_t\end{pmatrix} = \begin{pmatrix}0 \\ p_{22}\end{pmatrix} = \begin{pmatrix}0 \\ 0.8\end{pmatrix}$$

One shock at a time

- Set one ε to 1, keep the other at 0

Why s moves under ε_1 alone

- $p_{21} = 0.6$ passes ε_1 into u_s on impact

Responses to the first shock ε_1 alone

$\varepsilon_1 = 1$ and $\varepsilon_2 = 0$ today: the recursion starts from the first column of P

Impact	$\Delta r_t = 1$	$\Delta s_t = 0.6$
1 later	$\Delta r_{t+1} = 0.5 \times 1 = 0.5$	$\Delta s_{t+1} = -0.2 \times 1 + 0.4 \times 0.6 = 0.04$
2 later	$\Delta r_{t+2} = 0.5 \times 0.5 = 0.25$	$\Delta s_{t+2} = -0.2 \times 0.5 + 0.4 \times 0.04 = -0.084$

Horizon	Response of r to ε_1	Response of s to ε_1
Impact	1	0.6
1 period later	0.5	0.04
2 periods later	0.25	−0.084

Responses to the second shock ε_2 alone

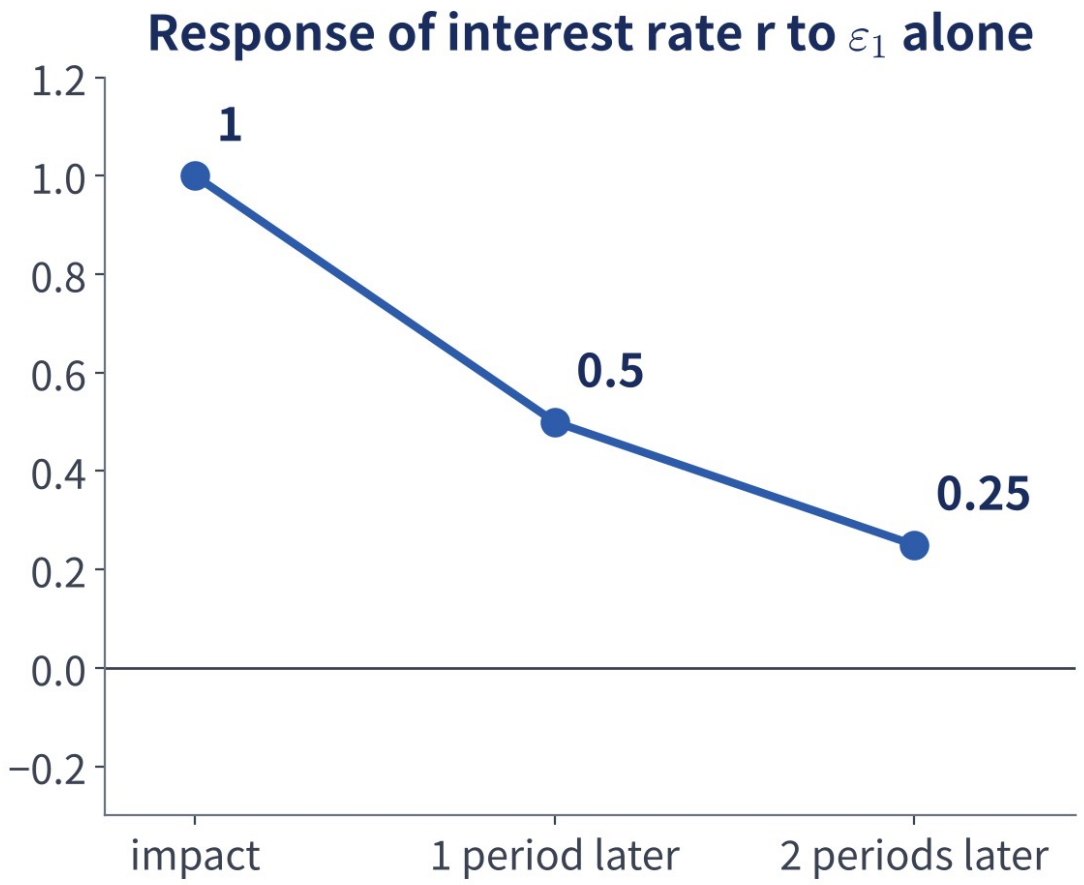
$\varepsilon_2 = 1$ and $\varepsilon_1 = 0$ today: start from P's second column; $a_{12} = 0$ keeps the rate at zero

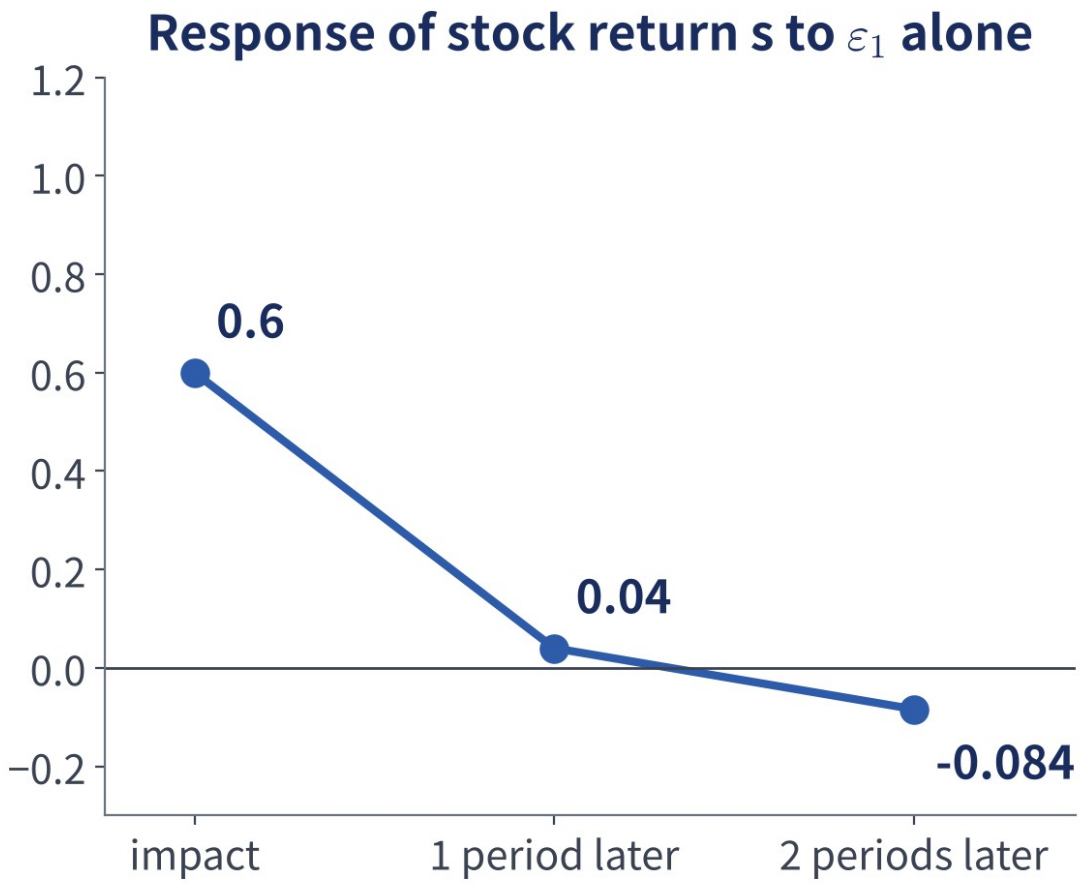
Impact	$\Delta r_t = 0$	$\Delta s_t = 0.8$
1 later	$\Delta r_{t+1} = 0.5 \times 0 = 0$	$\Delta s_{t+1} = -0.2 \times 0 + 0.4 \times 0.8 = 0.32$
2 later	$\Delta r_{t+2} = 0.5 \times 0 = 0$	$\Delta s_{t+2} = -0.2 \times 0 + 0.4 \times 0.32 = 0.128$

Horizon	Response of r to ε_2	Response of s to ε_2
Impact	0	0.8
1 period later	0	0.32
2 periods later	0	0.128

Impulse responses to the first shock ε_1 alone

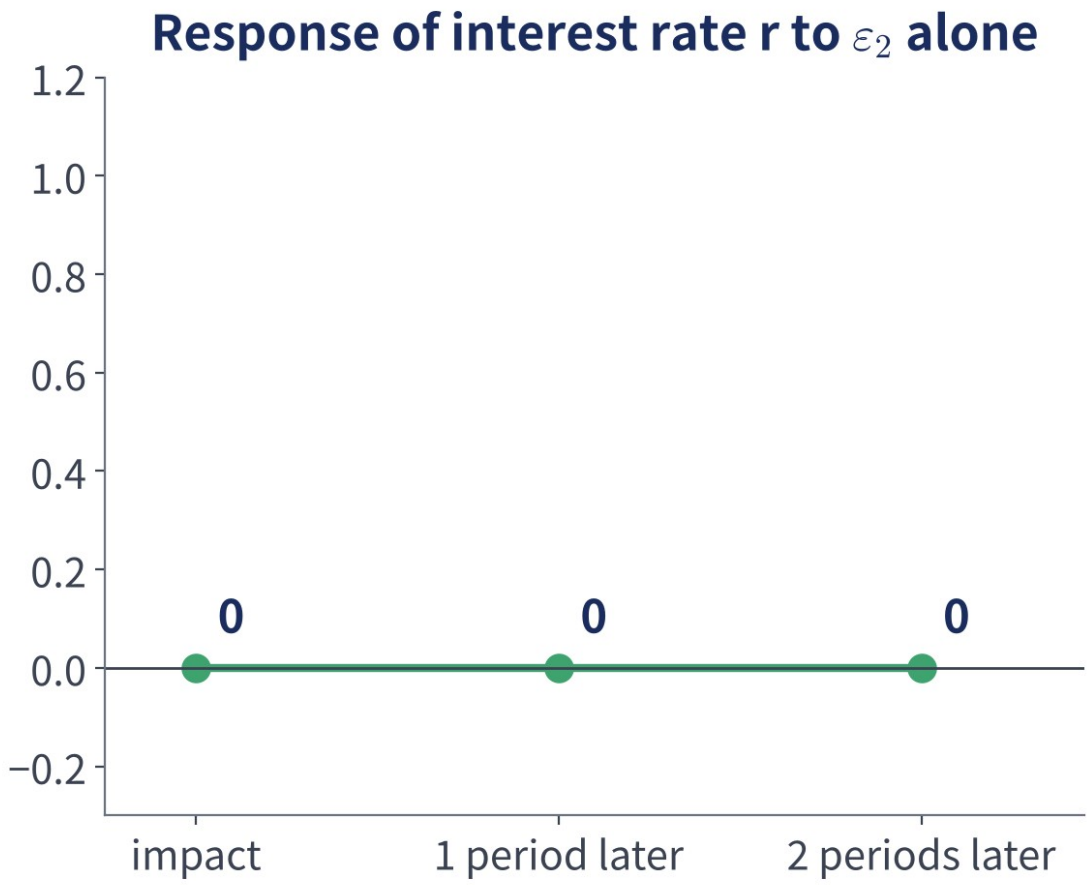
The rate decays 1 → 0.5 → 0.25; the return turns from +0.6 to −0.084 as the lagged rate effect arrives

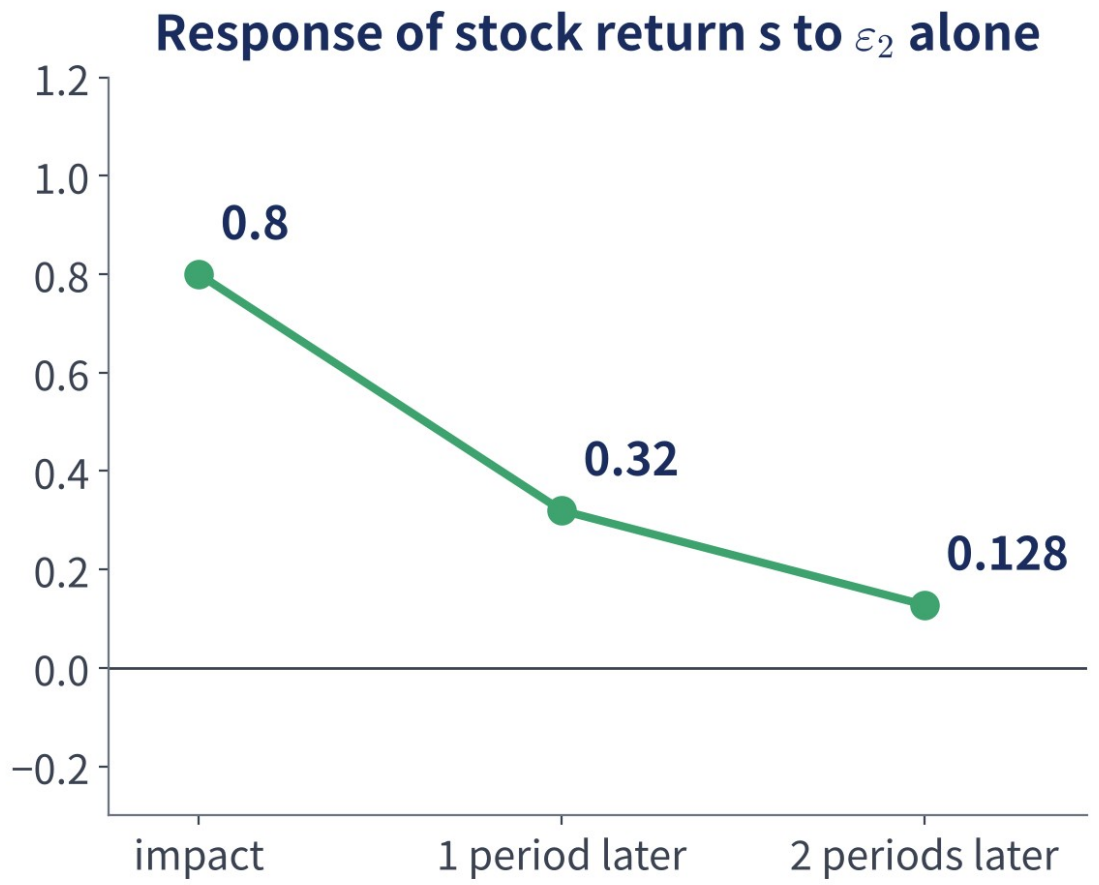




Impulse responses to the second shock ε_2 alone

The rate never moves ($p_{12} = 0$, $a_{12} = 0$); the return decays $0.8 \to 0.32 \to 0.128$





In matrix form the responses are P, AP and A^2P

Column j of each matrix is the response to a unit shock ε_j

$$\Psi_0 = P = \begin{pmatrix} 1 & 0 \\ 0.6 & 0.8 \end{pmatrix} \quad \Psi_1 = AP = \begin{pmatrix} 0.5 & 0 \\ 0.04 & 0.32 \end{pmatrix} \quad \Psi_2 = A^2P = \begin{pmatrix} 0.25 & 0 \\ -0.084 & 0.128 \end{pmatrix}$$

Division of labour

- P determines the initial impact
- A propagates it period by period
- Read direction, size and persistence

In applied work

- Inspect confidence intervals
- Return responses differ from cumulative price responses

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PART 5 · FEVD

Forecast Error Variance Decomposition

Decomposition tables for the stock return, one to three periods ahead

FEVD splits forecast uncertainty by shock

Forecast error variance decomposition: how much of the forecast error comes from each shock

The share of a variable's h-step-ahead forecast error variance that is due to each orthogonal shock.

1 · Square

Square each shock's IRF response

ups and downs both count

2 · Add

Add the squares up to the forecast horizon

h steps use h responses

3 · Divide

Divide each shock's sum by the total

shares sum to 100%

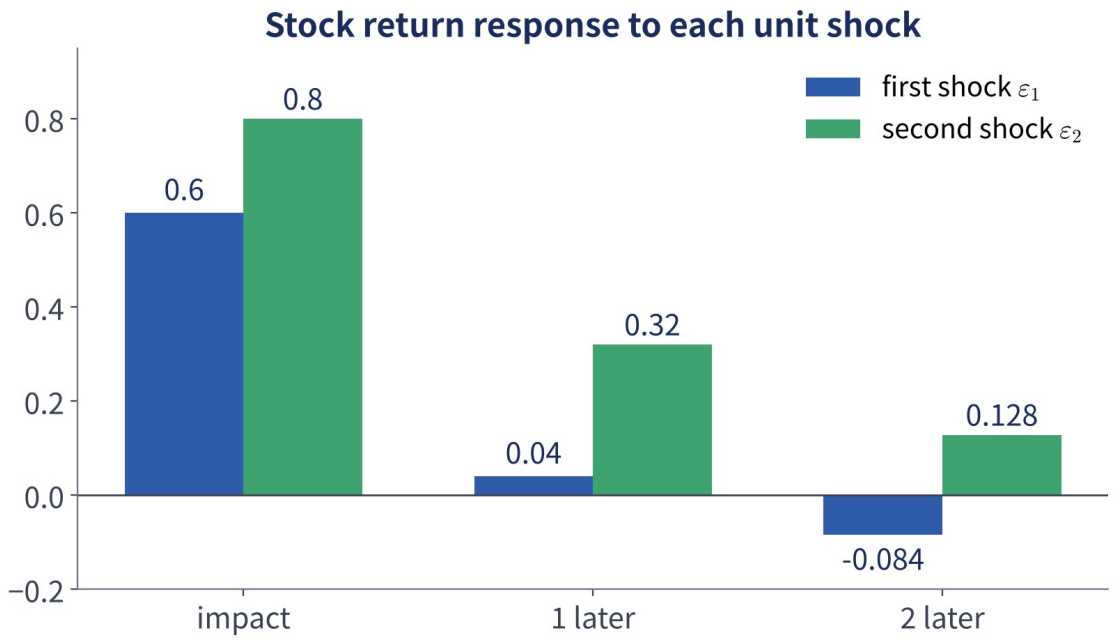
Because the shocks are uncorrelated, there are no cross terms: the pieces simply add up.

The inputs are the return responses to each shock

Taken from Part 4: each shock occurs alone with size one

Horizon	ε_1 alone	ε_2 alone
Impact	0.6	0.8
1 period later	0.04	0.32
2 periods later	−0.084	0.128

Each column is the stock return response when that shock occurs by itself.



Decomposing the stock return forecast error variance

Squared responses are added up to each horizon, then divided by the total

Horizon	From ε_1	From ε_2	Total	ε_1 share	ε_2 share
1 ahead	$0.6^2 = 0.36$	$0.8^2 = 0.64$	1.000	36.0%	64.0%
2 ahead	$0.36 + 0.04^2 = 0.3616$	$0.64 + 0.32^2 = 0.7424$	1.104	32.8%	67.2%
3 ahead	$0.3616 + 0.084^2 = 0.3687$	$0.7424 + 0.128^2 = 0.7588$	1.127	32.7%	67.3%

Reading the rows

- Each row adds one squared response

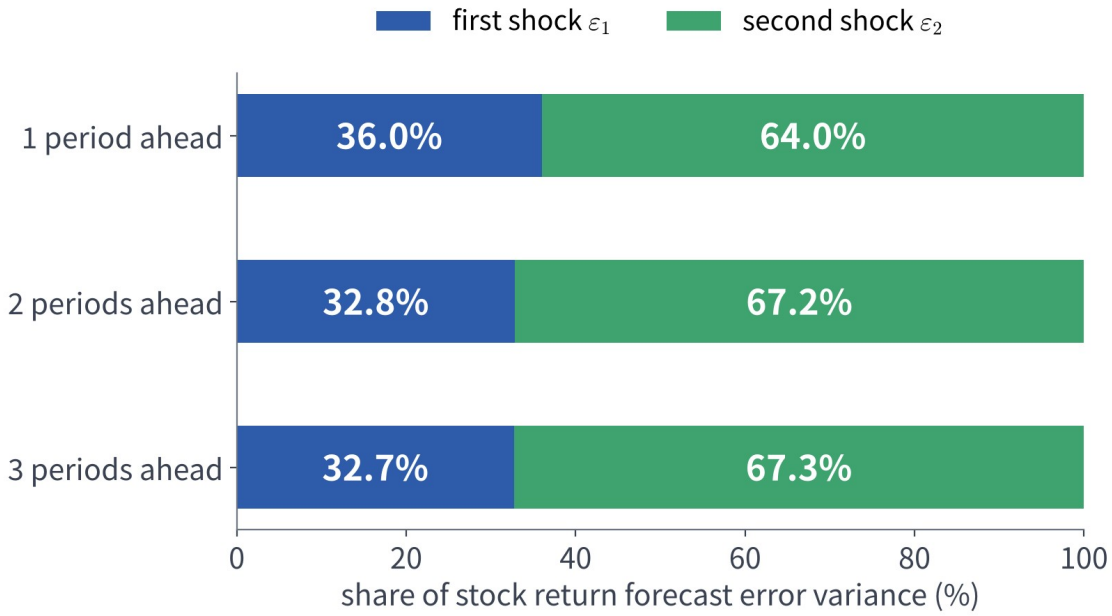
The 2-later response

- First enters at 3 ahead, not at 2 ahead

The FEVD table shows who drives each forecast

The format statistical software reports: shares of forecast error variance, in percent

Variable	Horizon	ε_1	ε_2
Interest rate	1, 2, 3 ahead	100%	0%
Stock return	1 ahead	36.0%	64.0%
Stock return	2 ahead	32.8%	67.2%
Stock return	3 ahead	32.7%	67.3%



The rate never responds to ε_2, so all of its forecast uncertainty comes from ε_1. These are shares of variance, not of the return itself.

An h-step FEVD uses the first h IRF responses

The shock that hits last acts on impact; the earliest one has propagated longest

FEVD horizon	IRF responses used
1 period ahead	Impact
2 periods ahead	Impact and 1 period later
3 periods ahead	Impact, 1 period later and 2 periods later

An IRF response at k periods later
first enters the FEVD at k + 1 periods ahead.

Granger, IRF and FEVD support different conclusions

Economic interpretation depends on the ordering and the model

Tool	Interpretation
Granger	Does past information add predictive power?
IRF	What changes relative to the no-shock path?
FEVD	What share of forecast error variance comes from each shock?

Cholesky creates uncorrelated shocks.
Ordering allocates the overlapping movements.

Three ideas carry the whole calculation

1

u is the original error; ε is the new shock

Cholesky turns correlated u into uncorrelated, unit-variance ε through $u = P\varepsilon$ and $\Sigma u = PP'$.

2

P sets initial effects; A propagates them

Responses are P, AP and A^2P: impact, one period later and two periods later.

3

IRF traces responses; FEVD squares them

Accumulated squared responses divided by their total give each shock's variance share.

Granger asks about prediction; IRF and FEVD ask about shocks under an ordering.

A

APPENDIX A

Cholesky Step by Step

Three equations, top-down solution, and the numerical check behind Part 3

Σu supplies three equations for three unknowns

Two mixing equations, three coefficients: p_{11}, p_{21} and p_{22}

$$\sigma_r^2 = \mathrm{Var}(p_{11}\varepsilon_1) = p_{11}^2$$
$$\sigma_{rs} = \mathrm{Cov}(p_{11}\varepsilon_1,\ p_{21}\varepsilon_1 + p_{22}\varepsilon_2) = p_{11}p_{21}$$
$$\sigma_s^2 = \mathrm{Var}(p_{21}\varepsilon_1 + p_{22}\varepsilon_2) = p_{21}^2 + p_{22}^2$$

Σu holds three distinct numbers: exactly enough for three unknowns.

What makes it work

- Rate variance gives p_{11}
- Covariance gives p_{21}
- Return variance gives p_{22}
- Unit shock variances used
- Cross term vanishes since $\mathrm{Cov}(\varepsilon_1, \varepsilon_2) = 0$

The factor is solved from the top down

Each step uses only the coefficients found in the step before

$$p_{11} = \sqrt{\sigma_r^2} \qquad p_{21} = \frac{\sigma_{rs}}{p_{11}} \qquad p_{22} = \sqrt{\sigma_s^2 - p_{21}^2}$$

$p_{12} = 0$

Triangular structure fixes one of the four entries of P

Var(ε) = 1

Unit variance for both new shocks

Cov = 0

The new shocks are uncorrelated

Diagonal > 0

Positive diagonal entries make P unique

Correlation 0.6 yields 0.8 on the second diagonal

Both original error variances equal one — a feature of this example, not a general requirement

$$\Sigma_u = \begin{pmatrix} 1 & 0.6 \\ 0.6 & 1 \end{pmatrix} \Rightarrow p_{11} = 1, \quad p_{21} = \frac{0.6}{1} = 0.6, \quad p_{22} = \sqrt{1 - 0.36} = 0.8$$

$$P = \begin{pmatrix} 1 & 0 \\ 0.6 & 0.8 \end{pmatrix} \qquad u_r = \varepsilon_1, \qquad u_s = 0.6\,\varepsilon_1 + 0.8\,\varepsilon_2$$

Why correlation and covariance coincide here

- Correlation = covariance / (SD × SD) = 0.6 / (1 × 1) = 0.6

Recovering ε: remove the overlap, then rescale

Define $v = u_s - 0.6u_r$: the part of the return error not shared with the rate error

$$\varepsilon_1 = u_r \qquad\qquad \varepsilon_2 = \frac{u_s - 0.6\,u_r}{0.8}$$

$$\begin{aligned} v &= u_s - 0.6\,u_r \\ \mathrm{Cov}(u_r, v) &= 0.6 - 0.6 \times 1 = 0 \\ \mathrm{Var}(v) &= 1 + 0.6^2 - 2 \times 0.6 \times 0.6 = 0.64 \\ \mathrm{SD}(v) = 0.8 \;&\Rightarrow\; \mathrm{Var}(v/0.8) = 1 \end{aligned}$$

Two moves

- Subtract the overlap → covariance 0
- Divide by SD 0.8 → variance 1

Uncorrelated shocks are constructed, not assumed

The same steps in matrix form

$$\varepsilon = P^{-1}u \qquad \mathrm{Var}(\varepsilon) = P^{-1}\,\Sigma_u\,(P^{-1})' = I$$

We do not assume the original errors are uncorrelated.
We transform them into new shocks that have that property.

Input	Transform	Output
Correlated u with covariance Σu	$\varepsilon = P^{-1}u$ with $\Sigma u = PP'$	Var(ε) = I: uncorrelated, unit variance